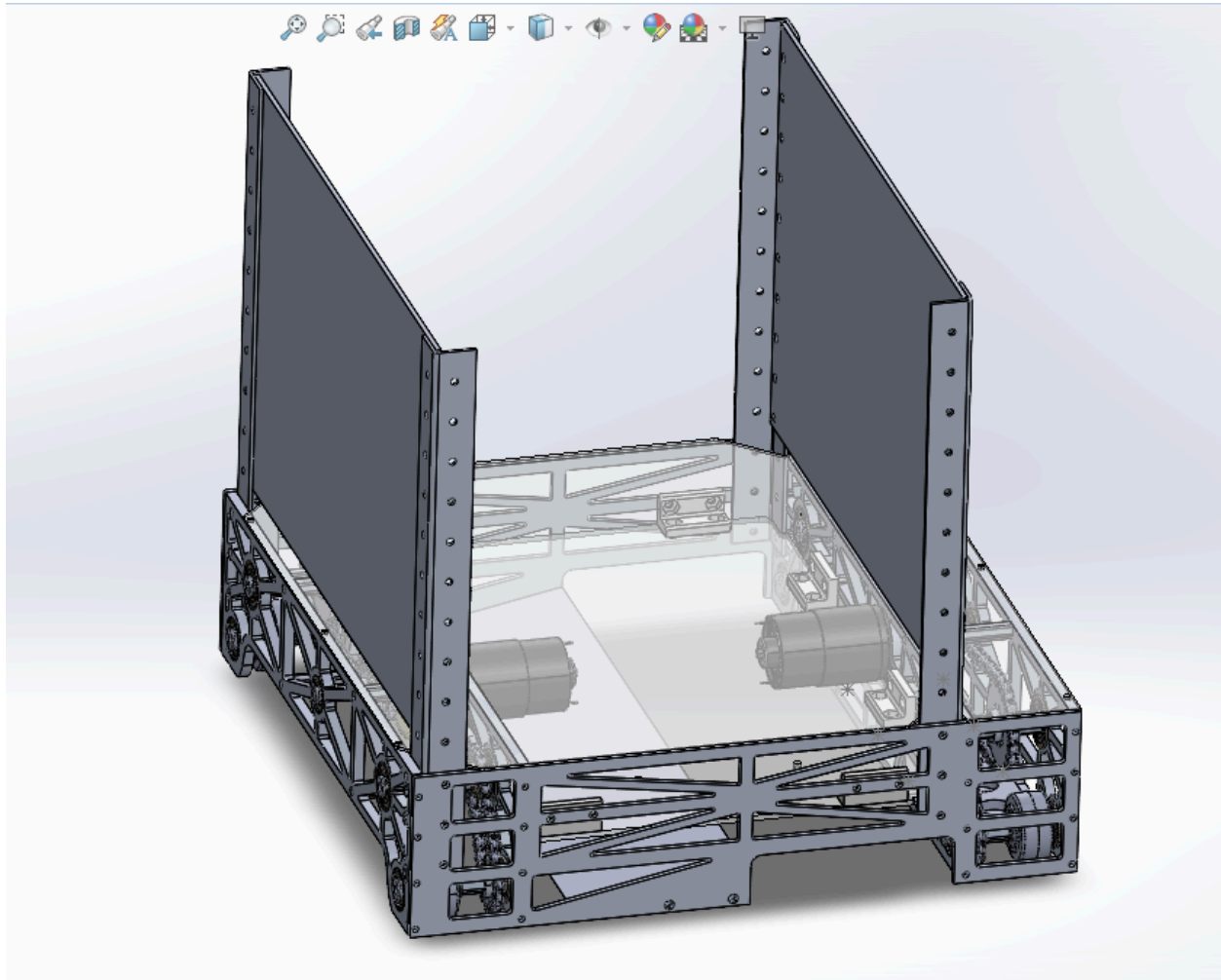
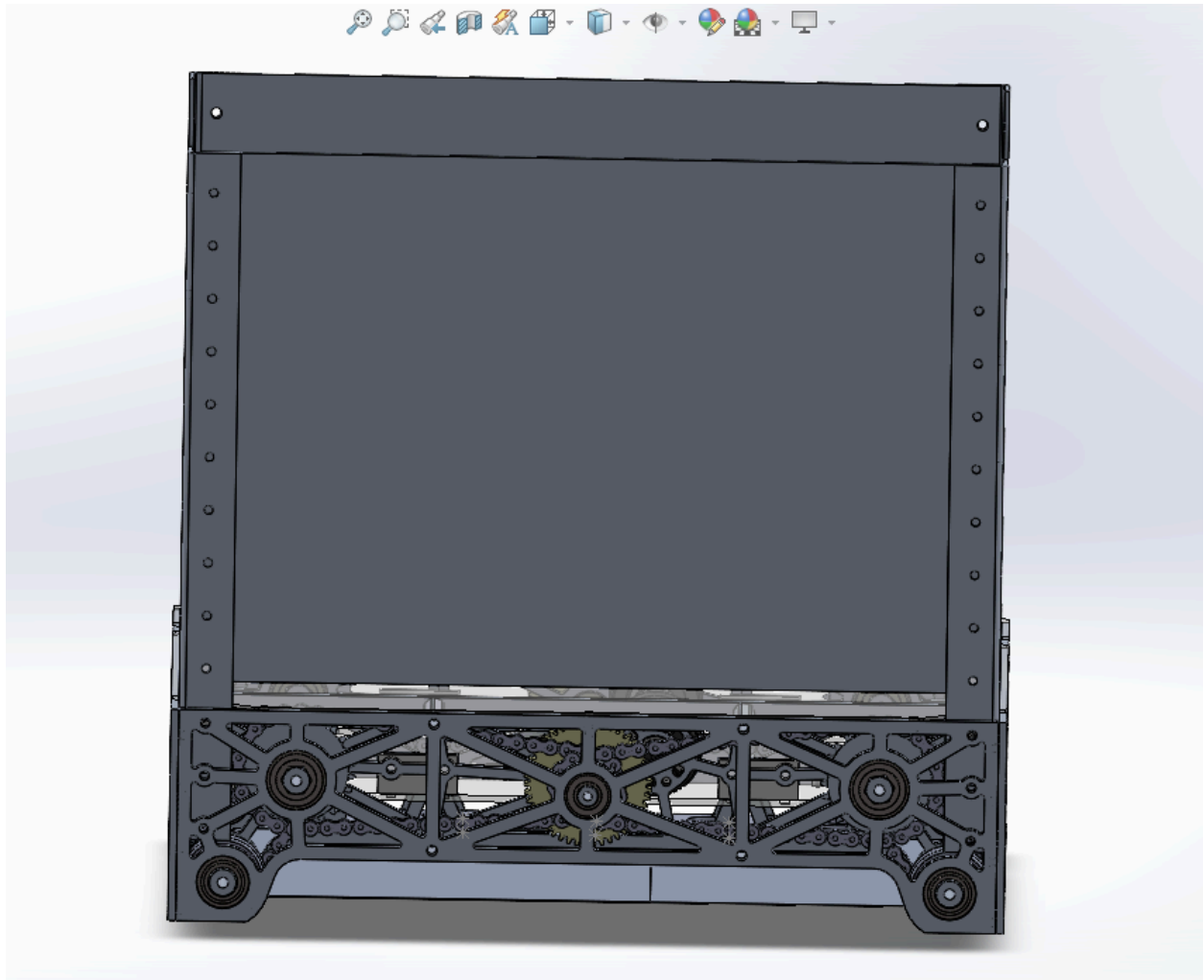


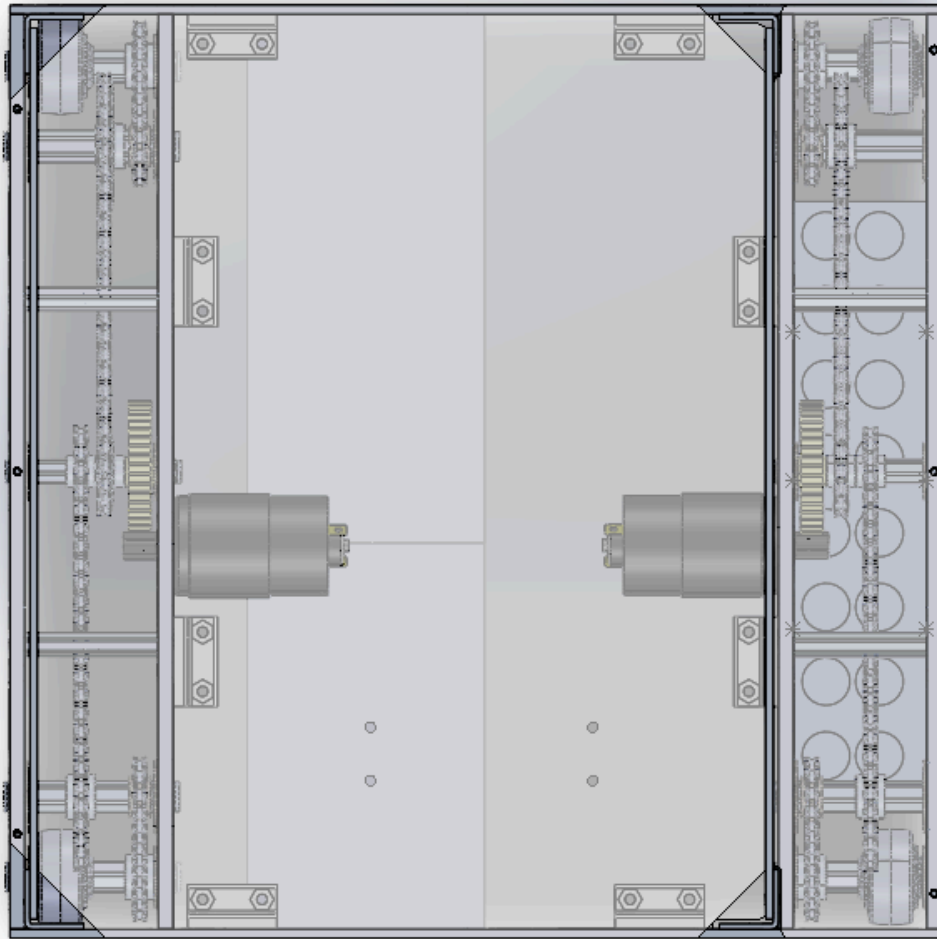
Kristina Sevier Paper Portfolio

ME 72 Robot Street Hockey Goalie Bot Design

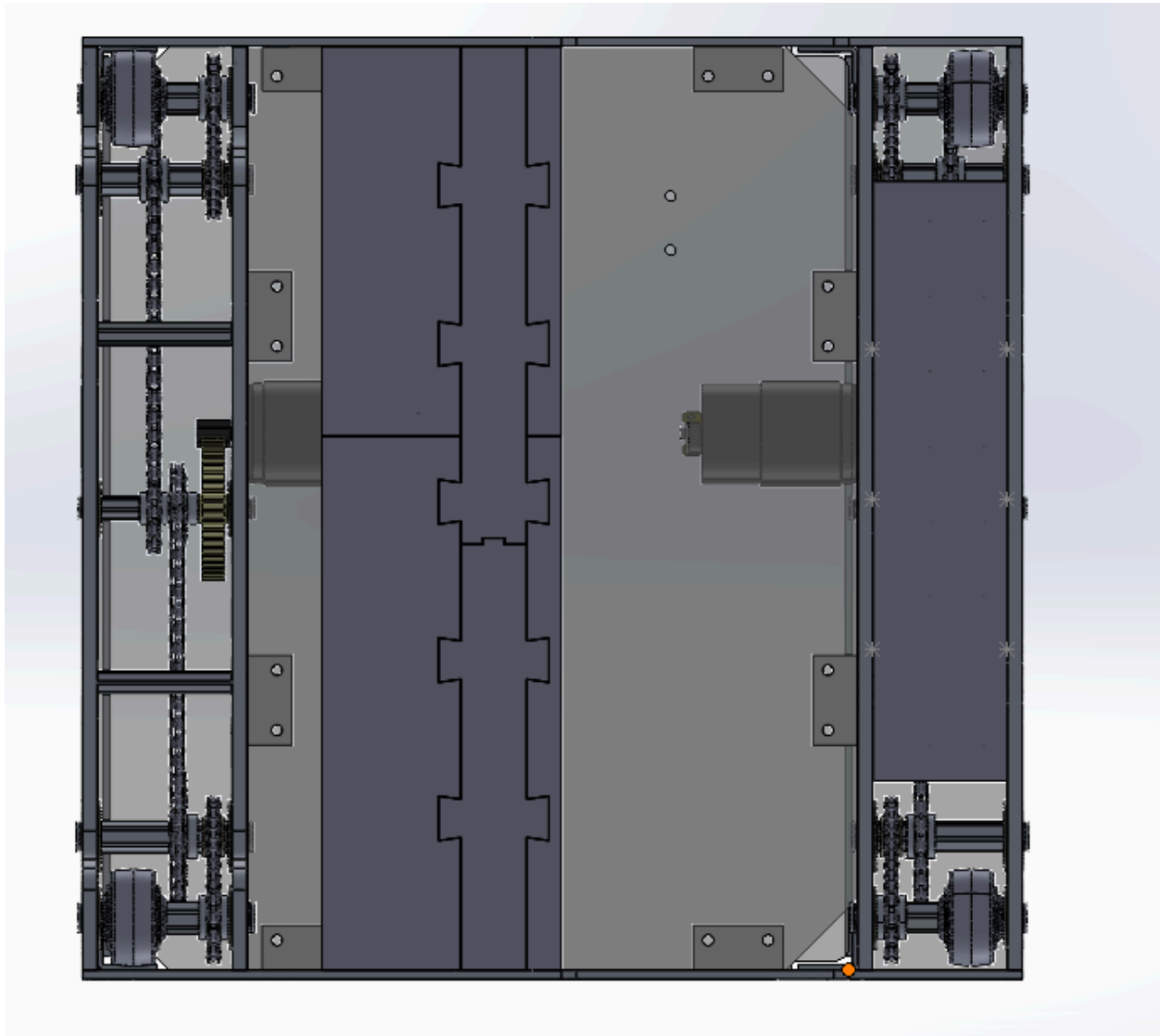




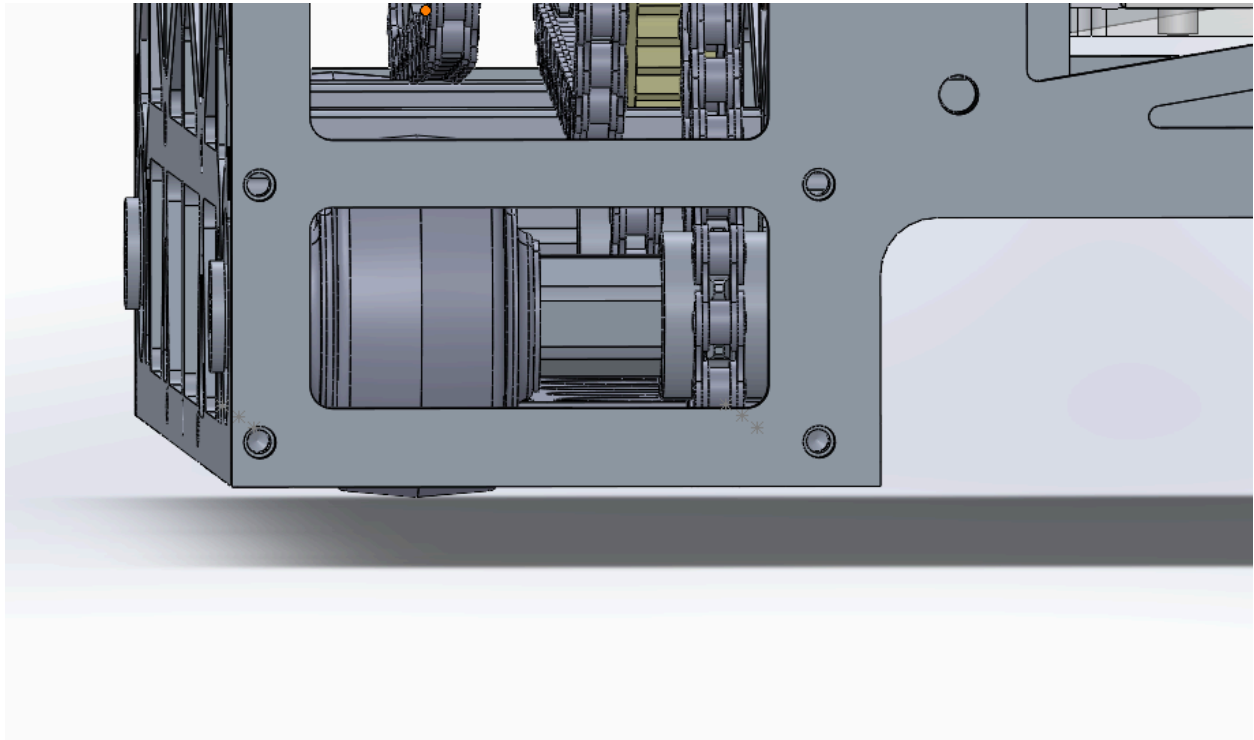
The top view of the goalie bot (no electronic implemented in the CAD)



Bottom view of the Goalie bot



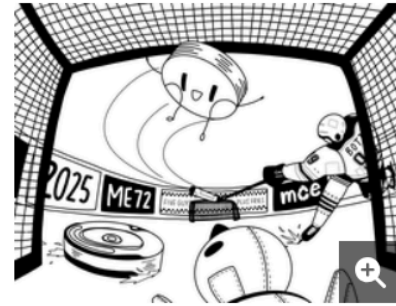
Close up showcasing how low the robot was to the ground



While the distance of ~ 0.1 inches between the robot and the wheel touching the ground theoretically could maximize durability and maneuverability, the floor was slightly uneven. I forgot to consider this at all, so sometimes our bot would struggle with maneuvering. We were still able to win the competition, but that is a mistake that I have learned from for the future.

THE MISSION

In the 40th Annual ME 072, teams of mechanical and civil engineering students will design and build a fleet of three remote-controlled (R/C) robots to compete in a high-energy Bot Hockey tournament. Inspired by traditional street hockey, this game challenges students to develop robots that can maneuver, pass, and shoot a regulation street hockey puck into their opponent's goal while defending their own. Unlike conventional Bot Hockey competitions, this year's game features a custom-designed 20 ft × 12 ft rink with a ferromagnetic surface, allowing robots to generate enhanced traction using magnets mounted on their chassis—a design element that introduces a unique mechanical and strategic challenge. The top-performing teams from the round-robin phase will advance to an elimination bracket, culminating in a championship match that determines the winner of this year's ME 072 Capstone Design Challenge.



COMPETITION RULES & TASKS

Each match features two teams of three robots competing in a five-minute game on the 20 ft × 12 ft magnetic rink. Robots must successfully shoot or push the puck into the opposing goal while leveraging increased traction from their magnetized chassis to execute high-precision movements, rapid accelerations, and defensive maneuvers. Teams must comply with strict size and weight constraints, requiring them to balance speed, agility, and durability in their designs. Physical contact is allowed and encouraged, provided it aligns with fair play and does not result in damage that cannot be repaired between matches. The game rewards precise control, strategic maneuvering, and effective use of autonomous or semi-autonomous functionalities. Sudden-death overtime or a shootout will determine the winner in the event of a tie.

DESIGN CHALLENGES

This competition provides students with a hands-on engineering experience that integrates mechanical design, electronics, coding, and advanced fabrication techniques. Each team must design and build three unique robots within a 16"×16"×16" footprint, adhering to a combined team weight limit of 45 lbs. A defining feature of this year's competition is the integration of magnetically enhanced traction, requiring teams to carefully engineer their robots to capitalize on increased normal force for stability, agility, and high-torque maneuvers.

Students will fabricate custom chassis and puck-handling mechanisms using additive manufacturing (3D printing) and precision machining in our student machine shop and maker space. Robots will incorporate custom electronics, actuators, and sensors, with control systems programmed using Arduino and Raspberry Pi platforms. Teams are encouraged to develop specialized puck shooters, enabling their robots to execute powerful and precise shots on goal, adding a dynamic offensive strategy to gameplay. Successful teams will optimize their drive systems for high-performance traction, engineer innovative puck control mechanisms, and integrate software-driven automation to dominate the rink. This competition challenges students to think like professional engineers, blending creativity with technical skill to bring their robotic athletes to life.